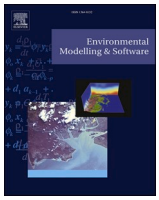
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Small increases in agent-based model complexity can result in large increases in required calibration data

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ABSTRACT

Agent-based models (ABMs) are widely used to analyze coupled natural and human systems. Descriptive models require careful calibration with observed data. However, ABMs are often not calibrated in a formal sense. Here we examine the impact of data record size and aggregation on the calibration of an ABM for housing abandonment in the presence of flood risk. Using a perfect model experiment, we examine (i) model calibration and (ii) the ability to distinguish a model with inter-agent interactions from one without. We show how limited data sets may not adequately constrain a model with just four parameters and relatively minimal interactions. We also illustrate how limited data can be insufficient to identify the correct model structure. As a result, many ABM-based inferences and projections rely strongly on prior distributions. This emphasizes the need for utilizing independent lines of evidence to select sound and informative priors.

1. Introduction

Agent-based models (ABMs) can be a useful tool for modeling and understanding how macro-scale/aggregate features of complex systems emerge from micro-scale/individual decisions, interactions, and feedbacks ("generative" social science (Epstein, 1999)). As a result, ABMs are used in many application areas, including land use change (Parker et al., 2003; Evans and Kelley, 2004; Brown et al., 2005; Evans and Kelley, 2008; Kelley and Evans, 2011; Evans et al., 2013; Brown et al., 2017; Dou et al., 2020; Li et al., 2020), ecology (Black and McKane, 2012; Grimm, 1999; van der Vaart et al., 2016), and climate change adaptation (Balbi et al., 2013; Barthel et al., 2008; Gerst et al., 2013; Schneider et al., 2000; Ziervogel et al., 2005; Lamperti et al., 2020). One key application is modeling flood risk and impacts due to the presence of observed socio-hydrological feedbacks (Aerts et al., 2018; Di Baldassarre et al., 2013). Flood risk modeling using ABMs is an extremely active area of research (Dubbelboer et al., 2017; Haer et al., 2016; Jenkins et al., 2017; Tonn and Guikema, 2017; Magliocca and Walls, 2018; de Koning and Filatova, 2019; Chandra-Putra and Andrews, 2020; O'Shea et al., 2020; Koc and Işık, 2020).

Models can be designed to address different questions about the modeled system, including prediction, explanation, and demonstration (Epstein, 2008). Bankes (Bankes, 1993) distinguishes consolidative models, which are used to simulate the dynamics of a well-understood system, from exploratory models, which are used to test hypotheses about model structures or system dynamics. A similar, but not identical, classification is proposed by Marks (Marks, 2011), who classifies models as demonstrative or descriptive. Demonstrative ABMs are used to illustrate that patterns of interest can be produced through local-level rules and interactions. Descriptive ABMs are intended to reproduce observed phenomena for the purpose of explanation, prediction, or both. The descriptive model category includes both simpler, "strategic" models, and more complex, "tactical" models (Holling and Watt, 1966). Early ABMs, such as the pioneering work on racial segregation (Schelling, 1971), were primarily demonstrative (Janssen and Ostrom, 2006; Marks, 2011). Over time, there has been an increase in the fraction of descriptive models (Janssen and Ostrom, 2006).

Both demonstrative and descriptive models require tests to ensure that the model works as intended. Descriptive models also benefit from a comparison of model output against observations (Marks, 2011). An example is pattern-based modeling (Grimm et al., 2005), in which one or more structural patterns in the data are compared to model output to see if the model is capable of simulating multi-scale aspects of the data. Another model checking procedure are simulations over a historical

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period to demonstrate that the model reproduces historical data, though this is not the same thing as demonstrating that the model is a reproduction of system dynamics (Oreskes et al., 1994), as all models are approximations of real processes (Box, 1976).

The observation that models are only capable of approximating, rather than reproducing real system dynamics (Box, 1976), shows the importance of model calibration for descriptive modeling. Calibration is the process of selecting model structures and parameter values (Oreskes et al., 1994). One common approach to ABM calibration is to tune model parameters until model outputs are close to the empirical data (Kelley and Evans, 2011; Schwarz and Ernst, 2009; van der Vaart et al., 2015). ABM calibration can be complicated by the path-dependence and feedbacks associated with many coupled natural-human systems. These path-dependencies and nonlinear dynamics can cause observed outcomes to be very sensitive to small changes in any part of the system. As a result, observations of these systems are highly contingent (Brown et al., 2005), and may reflect the influence of initial states or early shocks in a way which obscures the underlying dynamics. As a result, these procedures can lead to over-fitting the model to the calibration data. As data are merely one realization of an underlying data-generating and observation process, over-fitting data which may have been less probable than other observations could result in biased inferences and projections.

The risk of over-fitting may be particularly high when there are a large number of model parameters which have interactions, resulting in compensating effects. To account for the stochastic elements of a model, another approach to calibration is to choose a model structure and parameter values which are most probable given the observations and prior information about system dynamics (Jaynes, 2003; Robert, 2007). This approach, which we adopt here, is often called statistical calibration (Sansó and Forest, 2009; Lee et al., 2019). Statistical calibration involves using statistical inference to obtain parameter values or select model structures.

Another complication of ABM development is the risk of over-parameterization. Over-parameterized models include variables and dynamics outside of what is supported by the evidence. Due to the additional degrees of freedom, overly-parameterized models are more likely to overfit observations, and may generalize less well for the purposes of prediction while adding minimal theoretical benefit. However, whether descriptive ABMs are intended to be used for explanation or prediction, these features suggest a need for quantification of model and parametric uncertainty, as observed patterns may be dependent on external forcings (which might be stochastic) or initial conditions.

This discussion leads to a simple overarching question: How does the amount of data required to statistically calibrate agent-based models increase with model complexity? Increasing complexity of agent decision rules (in the sense of the number of parameters) and including agent-agent and agent-environment interactions and feedbacks (in the sense of emergence) can reduce the ability to constrain model parameters or test hypotheses, particularly when data may be relatively scarce.

We address this question using a Bayesian approach to uncertainty quantification. We focus on three specific questions. First, how do varying dataset sizes and structures affect the statistical calibration of an ABM? Second, can we assess the relative evidence for models with varying levels of complexity, either in terms of high-dimensional decision rules or the types of agent interactions with each other and the environment? Third, how are calibration and hypothesis-testing affected by the use of spatially-aggregated data (as opposed to observations of individual agents), which may be all that are available due to data-collecting limitations or considerations of anonymity?

In particular, we use a perfect model experiment. A perfect model experiment uses pseudo-observations generated from a known data-generating process, including known model parameters (see, *e.g.* (Reed and Kollat, 2012), or (Olson et al., 2013)). We generate pseudo-observations for a variety of spatial and temporal extents. We then calibrate the model using these pseudo-data (with different seeds, to approximate the more typical situation where the model and the "true" data-generating process are dissimilar). This gives us a best-case scenario for how much information these observations could provide about the considered model(s). Real observations would provide less information, given observational uncertainty and missing aspects of the model compared to the real data-generating process.

For a concrete example, we focus on the particular problem of modeling housing abandonment under flood risk, following Tonn and Guikema (Tonn and Guikema, 2017). Housing abandonment poses potentially severe economic problems for settlements along rivers and coastlines (Fowler et al., 2017). Indeed, recent work has shown that repetitive flooding can intensity outmigration, with an associated increase in climate gentrification (de Koning and Filatova, 2019). Residents who haven't experienced flooding themselves may abandon their homes if their neighbors do due to depreciating values or anticipation of future flooding. An associated ABM and a submodel with fewer interactions and feedbacks are illustrated by the influence diagrams in Fig. 1.

For the following work, we use relatively simple ABMs, with limited interactions between agents and their environment. This choice allows us to highlight some of the concepts and difficulties associated with the statistical calibration of even simple ABMs with relatively few parameters and feedbacks, with an understanding that these complications grow as models increase in complexity.

2. Methods

2.1. Models

We analyze two ABMs (represented by the influence diagram in Fig. 1). In both models, agents decide to vacate their homes using a probabilistic decision process (logistic regression), as opposed to maximizing utility or using heuristics (which are more common in ABMs in certain application areas, such as land use (Groeneveld et al., 2017)). Once a house is abandoned, there is a chance that it is occupied by a new agent in a subsequent year.

In the first considered model ("no-interactions"), the probability of housing abandonment is determined only by the frequency of experienced floods over the previous ten years. This model is parameterized using three parameters: the logistic regression intercept β_0, the logistic regression coefficient for flood frequency β_1, and the probability that vacant houses are filled by a new agent α. Denote by s_t^i the state of parcel i, where $s_t^i = 1$ if the parcel is occupied and 0 if the parcel is vacant. The probability that this parcel switches from occupied to abandoned is

$$p\left(s_t^i = 0 \middle| s_{t-1}^i = 1\right) = \text{logit}^{-1}(\beta_0 + \beta_1 r_t),$$

where r_t is the frequency of flooding events for the previous ten years. The probability that this parcel switches from abandoned to occupied is constant,

$$p\left(s_t^i = 1 \middle| s_{t-1}^i = 0\right) = \alpha.$$

In the second considered model ("spatial-interactions"), we include an additional logistic regression covariate, the fraction of neighboring plots which are vacant at the start of time t. As a result, this model has four parameters, including the coefficient for this neighboring-vacancy covariate. These state transition equations give both models a Markovian structure (see Supplementary Information S2.2).

Of course, these models could be further expanded, for example by including housing market dynamics (see Fig. 1). In this study, we focus for clarity just on the two models, as we are interested in understanding the process and results of statistical calibration for simple ABMs.

2.2. Data

We generated pseudo-observations for the perfect model experiment



Fig. 1. Influence diagrams of three nested ABMs for housing abandonment decisions under evolving flood pressure – The black components form a basic model without interactions (the "no interaction" model), in which abandonment decisions are based only on floods experienced by each agent. The blue and black components form a model with spatial interactions (the "spatial interactions" model), in which agents move due to experienced floods and the proportion of neighboring lots which are vacant.

using the model with spatial interactions, to see if we could successfully test for the spatial-interaction effect. The pseudo-observations are generated using the spatial-interactions model for an artificial riparian settlement and realizations of annual flood height maxima (see Supplemental Section S1). The parcel return periods and river heights are shown in Supplementary Figure S1. Parcel residency was initialized by assuming that each parcel had a 99% probability of having a resident at the start of the simulation. The additional dynamic mechanism resulting from spatial interactions leads to increased probabilities of parcel abandonment for all return periods across realizations of the stochastic process, even for parcels that are far from floods (Fig. 2).

As data may not be available in individualized forms, we examine the power of data for calibration and hypothesis testing about model structures in both individual and aggregate forms. In the individual case, the data set contains observations of each observed parcel at each time. In the aggregate case, we observe the total number of abandoned parcels at each time. As a result, the individually-resolved data consists of "maps" of occupancy state (whether each parcel is occupied or unoccupied), while the aggregated data is a time series of counts of unoccupied parcels.

2.3. Calibration

We use a Bayesian framework for model calibration, based on Bayes' Theorem (Bayes, 1763):

$$p(\theta|y) \propto p(y|\theta)p(\theta),$$

where $p(\theta|y)$ is the posterior density, $p(y|\theta)$ is the data likelihood, and $p(\theta)$ is the prior.

We constructed the priors that were used in this study (Table 1) using a rough understanding of the model dynamics, so that the resulting probabilities of abandonment seemed plausible. They are intentionally not centered on the known data-generating parameter values.

For both individual-parcel and aggregate data, we model the probability of each parcel being vacant and compute the appropriate likelihood, treating each parcel's vacant status at time t as independent and identically distributed conditional on the state (its vacancy status and, in the spatial-interactions model, that of its neighbors) in time $t-1$. This

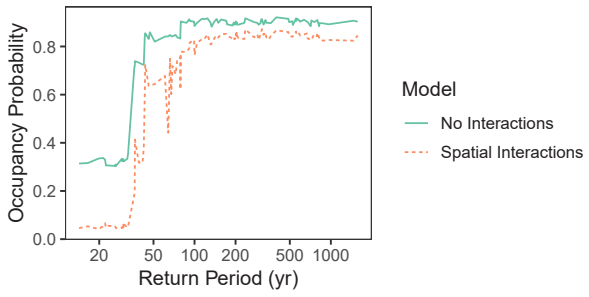


Fig. 2. Impact of spatial interactions on parcel occupancy probabilities – Occupancy probability (over 1000 sample realizations) for parcels with varying return periods after a 50-year model run for the no-interactions model and the spatial-interactions model. The simulated model used observations of 100 parcels.

Table 1
Prior distributions for the parameters in the no-interactions and spatial-interactions models.

Parameter	Prior Distribution
Intercept	Normal (-7, 1)
Flood Coefficient	Normal (19, 2)
Vacancy Coefficient	Normal (5, 2)
Vacancy Fill Probability	Beta(1, 10)

Markovian representation is common for many ABMs (Izquierdo et al., 2009), and we describe this process in Supplementary Section S2.2. In the individual data case, we use a binomial likelihood to model the probability of each parcel being vacant at each time. In the aggregate data case, we use a Poisson likelihood, which is commonly used to model count numbers, on the expected number of vacant parcels. For more details on how these likelihood functions are specified, see Supplemental Section S2.1.

The models described above are fast enough to use Markov Chain Monte Carlo (MCMC) for the Bayesian inversion. MCMC is an extremely general method for sampling from a target probability distribution, in this case the posterior distribution. The MCMC method constructs a Markov chain of samples which converge to the target distribution. MCMC has previously been used for calibrating ABMs (Keith and Spring 2013). There are many MCMC algorithms of varying complexity. We describe the algorithm we use, the Metropolis-Hastings algorithm (Metropolis et al., 1953; Hastings, 1970), in Supplementary Section S2.3. Other algorithms such as DREAM (Vrugt, 2016) or Hamiltonian Monte Carlo (Neal et al., 2011) are more sophisticated and powerful, but the relatively simple Metropolis-Hastings algorithm performs satisfactorily for these simple models, which have few parameters and have a runtime of under half a second for all considered cases. We use 150,000 Metropolis-Hastings iterations after a preliminary adaptive run (Vihola, 2012) of 30,000 iterations, which is used to estimate the starting value of the production run as well as the covariance matrix for the proposal distribution. The preliminary run is initialized at the maximum-likelihood estimate.

2.4. Model selection

We use Bayes factors to analyze the model selection problem (Kass and Raftery, 1995). Bayes factors compare the evidence for each model structure using a ratio of marginal likelihoods (see Supplementary Section S3.1). Intuitively, the marginal likelihood of a model captures the total probability assigned to the data by the calibrated model (across the entire posterior distribution). The Bayes factor then captures the extent to which one model better predicts the data. Much like other model comparison methods, such as the Akaike Information Criterion, or AIC (Akaike, 1974), Bayes factors penalize models which produce overly diffuse realizations (*e.g.* due to the presence of additional model parameters which are ill-constrained by calibration).

One important consideration when using Bayes factors is the dependence of marginal likelihoods on the choice of prior (Robert, 2007). From a Bayesian perspective, this is consistent with the idea that the Bayes factor is a metric comparing two model structures, as the selection of the prior is a key part of Bayesian model specification. Here,

we use the same priors for corresponding parameters to reduce this effect.

While the Bayes factor compares the ability of two models to reproduce the calibration data, a modeler may wish to select a model based on the expected ability of the model to predict out-of-sample data (Gelman et al., 2014). To this end, we also use an alternative metric, the Watanabe-Aikaike Information Criterion, or WAIC (Watanabe, 2010; Vehtari et al., 2017), which is described in more detail in Supplementary Section S3.2. However, in our case, WAIC may not provide a good approximation to the expected out-of-sample predicted error, as we do not calibrate our model repeatedly on shorter data subseries. We hence only use this metric to compare with the results of the Bayes factor analysis.

3. Results

3.1. Calibration

The structure of the data (individual-parcel versus spatially-aggregated) strongly influences the final shape of the posterior distribution, both due to the number of data points and the different likelihood function specifications. Fig. 3 shows the result of updating the prior distributions (specified in Table 1) with 50 years of pseudo-observations of 100 parcels. For certain key parameters (such as the logistic regression coefficient for the local flooding frequency), aggregated data (the total number of abandoned parcels at each time) leaves the posterior close to the prior (Fig. 3b). For individually-resolved data, the marginal posterior is sharpened much further (Fig. 3a).

While it appears from Fig. 3a that the original decision rules are not fully recovered (looking at the posterior density at the data-generating value), it is important to keep in mind the influence of stochasticity in the realized data. Running the same model with the same parameter values can yield model output with very different dynamics due to stochastic forcings, particularly in the presence of high levels of path dependence and positive feedbacks (see Fig. 4). Between the strong influence of the stochastic elements (the logistic regression) in the model and the relative lack of sensitivity to parameter values close to the data-generating value, it is not necessarily surprising that the data-generating value is assigned a relatively low density.

The full posterior parameter estimates feature a high degree of correlation between parameters (as seen in the pairs plot, Supplementary Figure S5). For example, the two logistic regression coefficients for flood frequency and proportion of neighboring abandoned parcels have a correlation coefficient of −0.66: a lower sensitivity to experienced floods can be offset by an increased sensitivity to neighbor behavior. Another example is the high positive correlation between the probability of a vacant lot being re-occupied and both the logistic regression intercept term and the coefficient for neighboring parcels (r = 0.73 in both cases). Similar interactions would be missed by a deterministic calibration combined with one-at-a-time sensitivity analysis (Ten Broeke et al., 2016).

We analyze the hindcasting ability of the posterior predictive distribution (shown in Fig. 5) to explore how the model was constrained by the observations and how well the model simulates the observed dynamics. The three-parameter no-interactions model is well constrained by smaller data sets. However, the poor fit of the posterior predictive distribution compared to the pseudo-observations for increased amounts

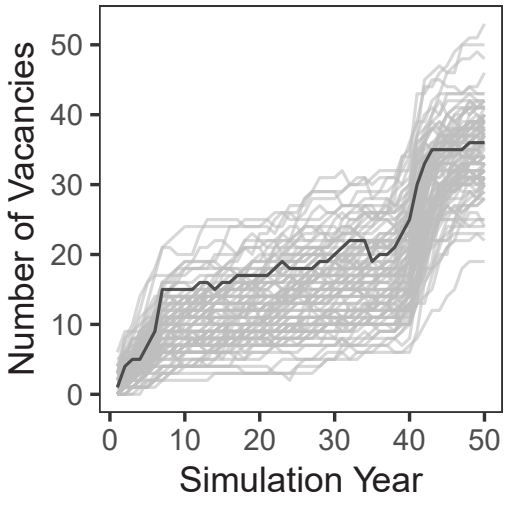


Fig. 4. Number of vacant parcels for one hundred sample realizations from the spatial-interactions model – The simulated model used observations of 100 parcels for 50 years. The black line is the realization used for that calibration experiment in this study.

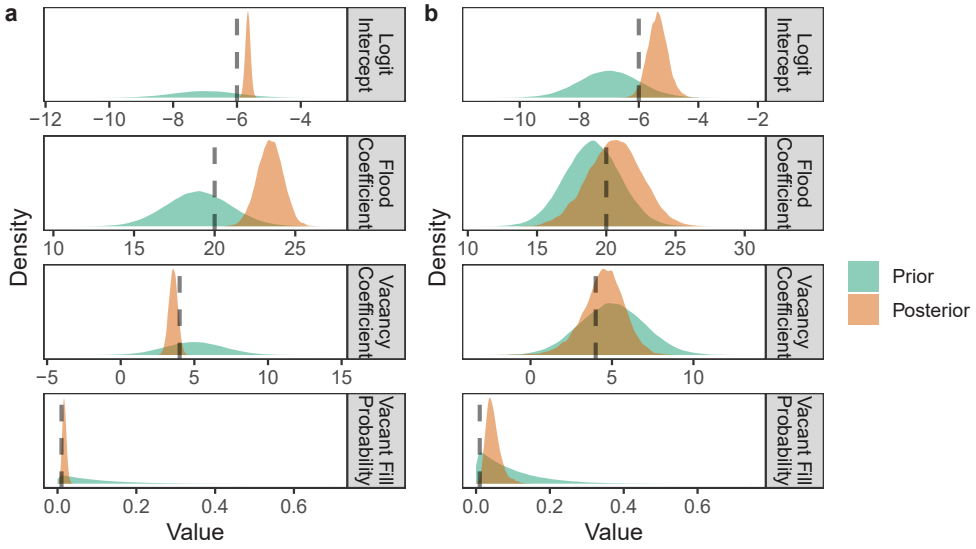


Fig. 3. Calibration results (prior and posterior distributions) for the spatial-interactions model after assimilating 50 years of observed data with 100 observed parcels – The dashed vertical line is the value used in the data-generating process. Panel a) is after assimilating individual-parcel data, and panel b) is after assimilating aggregated data.

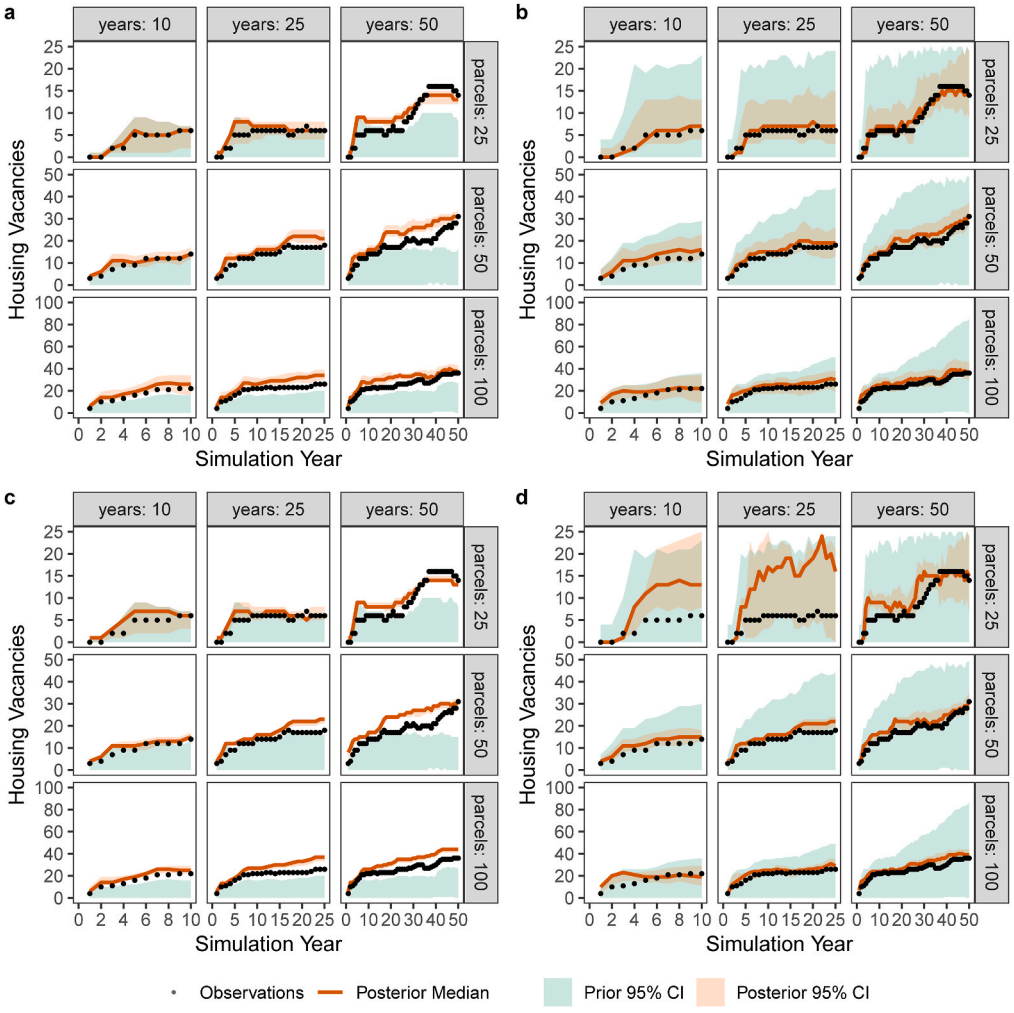


Fig. 5. Prior and posterior hindcasts of the number of vacant parcels – Panel a) shows the hindcasts for the no-interactions model with aggregated data, panel b) for the spatial-interactions model with aggregated data, panel c) for the no-interactions model with individual-parcel data, and panel d) for the spatial-interactions model with individual-parcel data. Each panel includes the hindcasts for varying combinations of observed years and parcels.

of data reveals the missing abandonment dynamic mechanism. Without the inclusion of spatial interactions, the no-interactions model calibration results in a higher sensitivity to experienced flooding to account for the data, which results in biased estimates of the number of abandoned parcels in later years. Meanwhile, the spatial-interactions model, which has one additional parameter and includes an interaction, requires more data to constrain the model (25 observed parcels is insufficient with up to 50 years of data), but, once constrained, fits the pseudo-observations better than the no-interactions model. In general, having a larger spatial domain/numbers of agents facilitates calibration more than having a longer data record. This pattern remains in our study when we calibrate the model with alternative pseudo-observations (see Supplemental Figures S7-S10).

While Fig. 5 might appear to show that calibration with aggregate data results in a better fit to the observations, we hypothesize that this is an artifact of two different components of the modeling process. First, the likelihood function used for the aggregate data calibration is different than the individual-data calibration (Poisson vs. binomial), which will change the shape of the posterior distribution. Second, in the aggregate case, we calibrate the model directly against the aggregated counts shown in Fig. 5, while in the individual-data case, the expected state of each parcel was used. These two differences make it difficult to compare the quality of the hindcast, as they are structurally different calibration procedures. We would expect that in the latter case, there is greater uncertainty about the total count of abandoned parcels. However, in some of the alternate realizations shown in Supplemental Figures S7-S10, using individually-resolved data can result in overconfidence, while, as seen in Supplemental Figure S6, using aggregated data resulted in more consistent parameter estimates. These details are likely to be extremely sensitive to the particular model specification.

3.2. *Model selection*

More complex ABMs can be thought of as being constructed by adding new interactions and feedbacks to simpler ABMs, as illustrated in Fig. 1. This allows us to view this type of model selection as hypothesis testing for the presence of additional feedback mechanisms (Cottineau et al., 2015) as well as potential over-parameterization.

Posterior model structural probabilities are calculated by combining prior knowledge and beliefs about the relative probability of the competing models with the Bayes factor. This prior information about the probability of various model structures can be guided by domain-specific insights that are gleaned, for example from other observations or other qualitative information (Forrester, 1980). In this stylized experiment, we make the prior assumption that the models with and without spatial-interactions are equally probable.

For our perfect model experiment, one would expect additional (in terms of the number of observations) and spatially explicit (rather than aggregated) data to improve the ability to distinguish between the data-generating spatial-interactions model and the simpler no-interactions

model, which is seen in Fig. 6. We neglect the case with 25 observed parcels due to unreasonably high estimates, likely due to the ill-constrained spatial-interactions model (however, the hindcasts in Fig. 5 shows the qualitatively better fit of the no-interactions model for this data set, particularly in the individual-data scenario). For individual-parcel data, with more than 25 observed parcels, there is at least strong evidence for the spatial-interactions model no matter how long the parcels were observed, which confirms the qualitative assessment (on the summary statistic of total abandoned parcels) obtained by comparing the hindcasts in Fig. 5c and d.

On the other hand, when aggregated data is used for calibration, there is essentially no quantitative evidence for the spatial-interactions model. This is the case whether we compare the models using Bayes factors or a predictive information criterion such as WAIC. The one-standard error range of the difference in WAIC between the spatial-interactions and the no-interactions model is between −2 and 2, which can be interpreted as no difference in support between the two models (Burnham and Anderson, 2004). However, a qualitative assessment obtained by comparing Fig. 5a and b might lead a researcher to conclude that the spatial-interactions model fits the observations better than the no-interactions model. This suggests that hindcasting can serve an important supporting role to quantitative model selection.

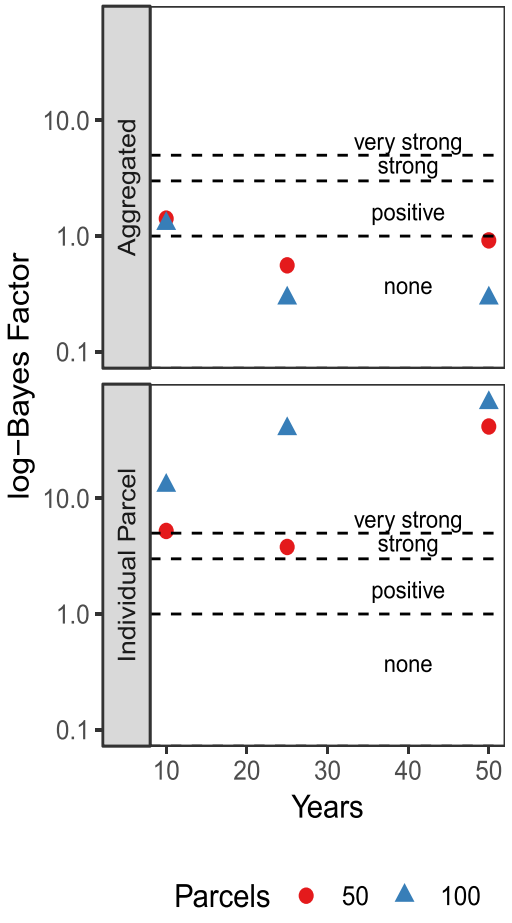


Fig. 6. Impact of data size and aggregation on relative model evidence – Log-Bayes Factors when comparing the with-interactions model to the no-interactions model for 50 and 100 parcels and 10, 25, and 50 years. The thresholds for varying degrees of evidence are taken from Kass and Raftery (Kass and Raftery, 1995). The marginal likelihoods for each model were estimated using bridge sampling (Meng and Wing, 1996) with 5000 samples and a truncated multivariate normal importance density.

4. Discussion

Statistical calibration is an important component of the descriptive agent-based modeling process due to the path-dependence associated with complex data-generating processes (as illustrated in Fig. 4). Using a statistical approach reduces the potential for over-fitting the model to the data, which is a risk with a deterministic approach to calibration (either through hand-tuning the parameters or through the use of an optimization routine). As a result, the calibration procedure can take into account that the observations may not have been the most likely outcomes given the understanding of the system dynamics (including the specifications of the likelihood function and prior distribution) that the model encodes, which has implications for parameter inference and model selection, and hence for future projections.

Our use of a didactic perfect model experiment somewhat limits the direct generalization of our results to other experiments using real-world data. In contrast to our perfect-model didactic example, the real-world data may be more noisy and will be generated through a process which is incompletely represented by any model (Box et al., 1979). In this case, statistical approaches to calibration are potentially more useful than in our experiment, as we can account for systematic model discrepancies (Kennedy and O'Hagan, 2001) through the construction of a likelihood which separates these discrepancy terms from white noise. As this perfect model experiment provides a somewhat best-case scenario for parameter inference and model selection, the demonstration of how limited data sets are unable to meaningfully constrain model hindcasts is likely to generalize to other studies. It is likely, however, that model selection as conducted in this study will be more complicated in the case where two imperfect (technically "wrong" (Box et al., 1979)) models are being compared, showing the importance of carefully synthesizing the available evidence, qualitative and quantitative, to provide justifications for a particular set of model features.

It is important to point out that different modeling aims would induce different calibration needs, with implications for the appropriateness of specific methods. Some users of demonstrative models, for example, might want to conduct a model-based exploratory analysis rather than making projections of future system outcomes. For example, one might explore how certain bottom-up rules which are broadly consistent with observations or other evidence could produce certain dynamics or outcomes. In this case, manual calibration based on some observational or survey data (for example) could be sufficient. As another example, a modeling exercise might be partially or entirely motivated by qualitative data which demonstrates that behaviors or outcomes follow certain patterns, in which case quantitative approaches to calibration are either inappropriate or should be modified to account for this additional information. If these aims are clearly indicated, the resulting data needs and calibration procedures can be reconciled with them.

While our chosen method of Metropolis-Hastings MCMC has the ability to produce high-fidelity approximations to the full joint probability distribution of the model parameters (Robert, 2014), it may be computationally intractable for complex models featuring long runtimes or high-dimensional parameter spaces. In this case, the relative simplicity of the models facilitated the use of MCMC. An additional complication is the need to specify a statistical likelihood function, which may be difficult for particular applications. In general, there is a trade-off between computational speed and accuracy of the resulting parameter distributions. Some other approaches to statistical calibration of ABMs, which are aimed at reducing computational requirements or likelihood specification, include statistical emulation (Lamperti et al., 2018; Oyebamiji et al., 2017), particle filtering (Kattwinkel and Reichert, 2017), and approximate Bayesian computation (Fabretti, 2018; Sirén et al., 2018; van der Vaart et al., 2015; van der Vaart et al., 2016). While these methods reduce the computational burden, they come at a cost of potentially severe statistical approximations that can influence the parameter estimates (Frazier et al., 2018; Künsch, 2013; Robert

et al., 2011; Singh et al., 2018).

Difficulties constructing likelihood functions may also emerge when a pattern-oriented approach towards model evaluation and selection is adopted, as it may be difficult to link multiple patterns together using an appropriate likelihood function. In these cases, alternative approaches like approximate Bayesian computation, or ABC (Sunnåker et al., 2013) would be useful. ABC may be particularly appropriate for pattern-oriented calibration and evaluation, though care should be taken that the set of statistics used to capture patterns is sufficient, as otherwise the approximation to the true posterior will be degraded.

An additional concern is the specification of prior distributions. When less data is available, the data will have less power to update the prior distributions. As we noted in the introduction, real data, as opposed to the pseudo-data used in this study, will also contain less information due to observational errors and non-modeled aspects of the underlying real stochastic process. This suggests that priors should be as informative as possible (with a strong warning that priors ought not to be more informative that can be supported). Golchi (Golchi, 2019) provides an overview of the usefulness of informative priors to solve several types of inference problems.

While we did not take prior correlations between parameters into account for this experiment, good priors for real-world problems will include prior information about correlations between parameters. Sound priors are informed by domain-specific insights, which may be obtained by, for example, expert assessments or survey results. One approach to creating informed priors which include information about the relationships between parameters is probabilistic inversion (Kraan and Cooke, 2000; Fuller et al., 2017), in which expert assessments (or, as an alternative, the results of judgement and decision-making or economic experiments) can be used to update more generic priors in a way which is consistent with those assessments or experimental results. This allows the survey or experimental participants to provide information directly about outcomes rather than about model parameters, and allows for a separation of the data involved in the prior construction and Bayesian updating processes to avoid information reuse.

As our results illustrate, each additional parameter or interaction in an ABM can considerably increase the calibration data requirements. Trying to include every hypothesized feedback mechanism in the final model choice, without supporting evidence, can pose problems from statistical as well as decision-theoretical points of view (Box, 1976; Jaynes, 2003; Robert, 2007). Starting with a simple model and adding complexity when supported by the data can produce more skillful hindcasts, projections, and more powerful insights (Box et al., 1979; Holling and Watt, 1966). Indeed, our analysis demonstrates that increased model complexity can result in an increased data need. While our simple models were motivated by the problem of housing abandonment in the presence of flood risks, we see no reason why this general point would not be applicable more broadly, though the details of the analysis would differ depending on the complexity of the model structure.

Author contributions statement

V.S. and K.K. conceptualized the research. V.S. wrote the model and analysis codes. V.S. and K.K. designed the figures and wrote the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envsoft.2021.104978.

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